

Math Class 11 Federal Board

Chapter # 8 Trigonometry

MOST IMPORTANT Questions

Example 18: Show that: $\sin 70^\circ \sin 30^\circ \cos 20^\circ \cos 10^\circ = \frac{1}{4} \cos 10^\circ + \frac{\sqrt{3}}{16} + \frac{1}{8} \sin 40^\circ$

$$\text{L.H.S} = \sin 70^\circ \sin 30^\circ \cos 20^\circ \cos 10^\circ$$

$$= \frac{1}{2} \sin 70^\circ \cos 20^\circ \cos 10^\circ$$

$$= \frac{1}{2} \times \frac{1}{2} (2 \sin 70^\circ \cos 20^\circ) \cos 10^\circ$$

$$= \frac{1}{4} [\sin 90^\circ + \sin 50^\circ] \cos 10^\circ$$

$$= \frac{1}{4} (1) \cos 10^\circ + \frac{1}{4} \sin 50^\circ \cos 10^\circ$$

$$= \frac{1}{4} \cos 10^\circ + \frac{1}{4} \times \frac{1}{2} (2 \sin 50^\circ \cos 10^\circ)$$

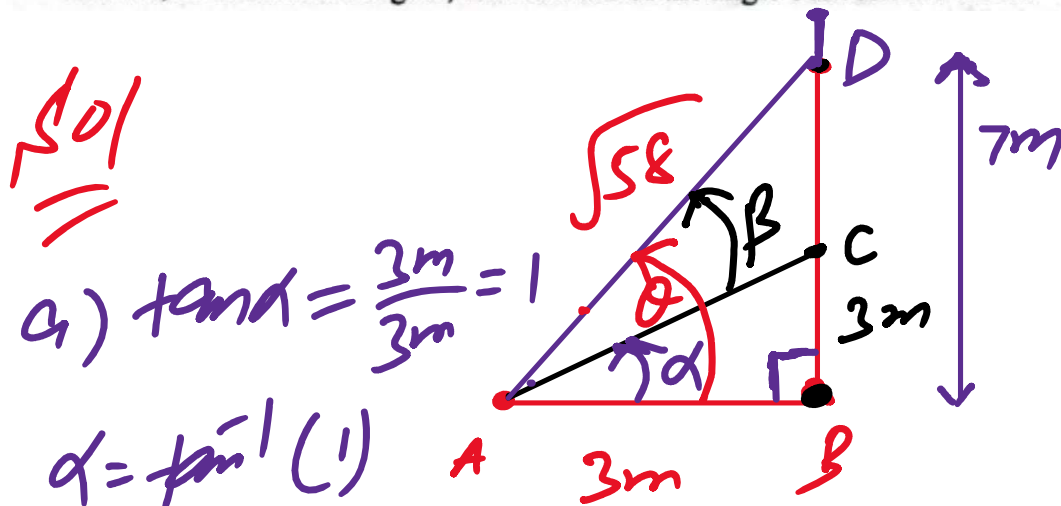
$$= \frac{1}{4} \cos 10^\circ + \frac{1}{8} [\sin 60^\circ + \sin 40^\circ]$$

$$= \frac{1}{4} \cos 10^\circ + \frac{1}{8} \frac{\sqrt{3}}{2} + \frac{1}{8} \sin 40^\circ$$

$$= \frac{1}{4} \cos 10^\circ + \frac{\sqrt{3}}{10} + \frac{1}{8} \sin 40^\circ$$

14. A telephone pole is braced by two wires that are both fastened to the ground at a point 3m from the base of the pole. The shorter wire is fastened to the pole 3m above the ground and the longer wire 7m above the ground.

- What is the measure, in degrees, of the angle that the shorter wire makes with the ground?
- Let θ be the measure of the angle that the longer wire makes with the ground. Find $\sin \theta$ and $\cos \theta$.
- Find the cosine of the angle between the wires where they meet at the ground.
- Find, to the nearest degree, the measure of the angle between the wires.



$$\boxed{\alpha = 45^\circ}$$

b) $\sin \theta = ?$

$$|AD|^2 = |AB|^2 + |BD|^2$$

$$AD = \sqrt{9 + 49} = \sqrt{58}$$

$$\boxed{\sin \theta = \frac{7}{\sqrt{58}}, \quad \cos \theta = \frac{3}{\sqrt{58}}}$$

c) Let β be the angle b/w two wires,

$$\beta = \theta - \alpha = \theta - 45^\circ$$

(... 10 - 45°)

$$\begin{aligned}\cos \beta &= \cos(\theta - 45^\circ) \\ &= \cos \theta \cos 45^\circ + \sin \theta \sin 45^\circ \\ &= \frac{3}{\sqrt{38}} \cdot \frac{1}{\sqrt{2}} + \frac{7}{\sqrt{58}} \times \frac{1}{\sqrt{2}}\end{aligned}$$

$$= \frac{3}{\sqrt{116}} + \frac{7}{\sqrt{116}}$$

$$\cos \beta = \frac{10}{\sqrt{116}} \Rightarrow \boxed{\cos \beta = 0.9285}$$

$$\theta) \quad \beta = \cos^{-1}(0.9285)$$

$$\checkmark \quad \boxed{\beta = 21.8^\circ \approx 22^\circ}$$

or

$$\beta = \theta - \alpha = \theta - 45^\circ \rightarrow \text{CC}$$

$$\beta = \theta - \alpha = 45^\circ \rightarrow (1)$$

$$\tan \theta = \frac{7}{3} \Rightarrow \theta = \tan^{-1}\left(\frac{7}{3}\right)$$

$$\boxed{\theta = 66.8^\circ}$$

$$\beta = 66.8^\circ - 45^\circ = 21.8^\circ$$

$$\Rightarrow \boxed{\beta \approx 22^\circ}$$